DoRA: Weight-Decomposed Low-Rank Adaptation

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Background

Problem

Full Finetuning of LLMs on all of their parameters is computationally too expensive.

Low rank assumption

Interpretate a LLM as a sequence of weight matrices. We assume that the optimal update ΔW of a $m \times n$ weight matrix W has low rank $l \ll m$.

LoRA Algorithm

Finetune LLM by training

$$\Delta W = A \cdot B$$

Such that A is a $m \times l$ and B is a $l \times n$ matrix.

Empirical Observation

Accuracy gap between LoRA and Full Finetuning

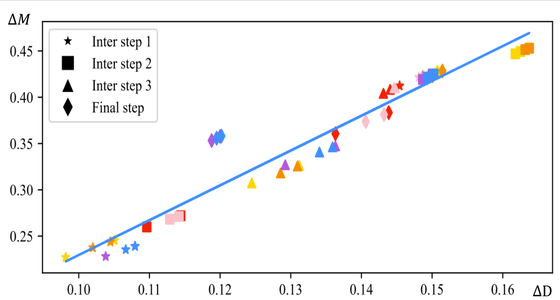
Weight and Direction of a Matrix

Given a $m \times n$ matrix W, let

- m be the $n \times 1$ magnitude vector containing the norm of the column vectors of W
- W_0 be the direction matrix with normalized columns

Failure of LoRA

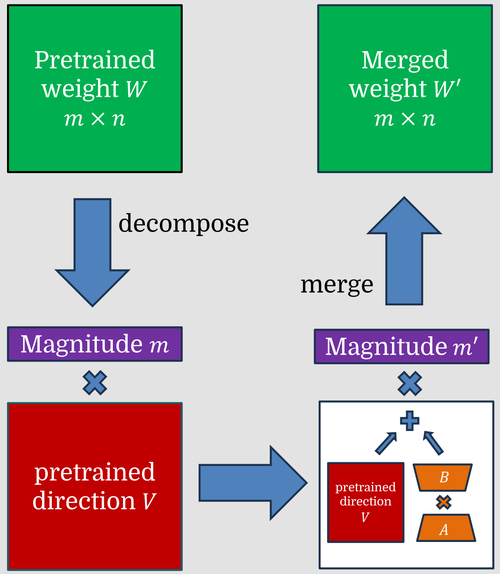
magnitude is linearly dependent on direction.



Methods

DoRA

Idea: Train magnitude and direction separately



Implementation

Pretrained Models

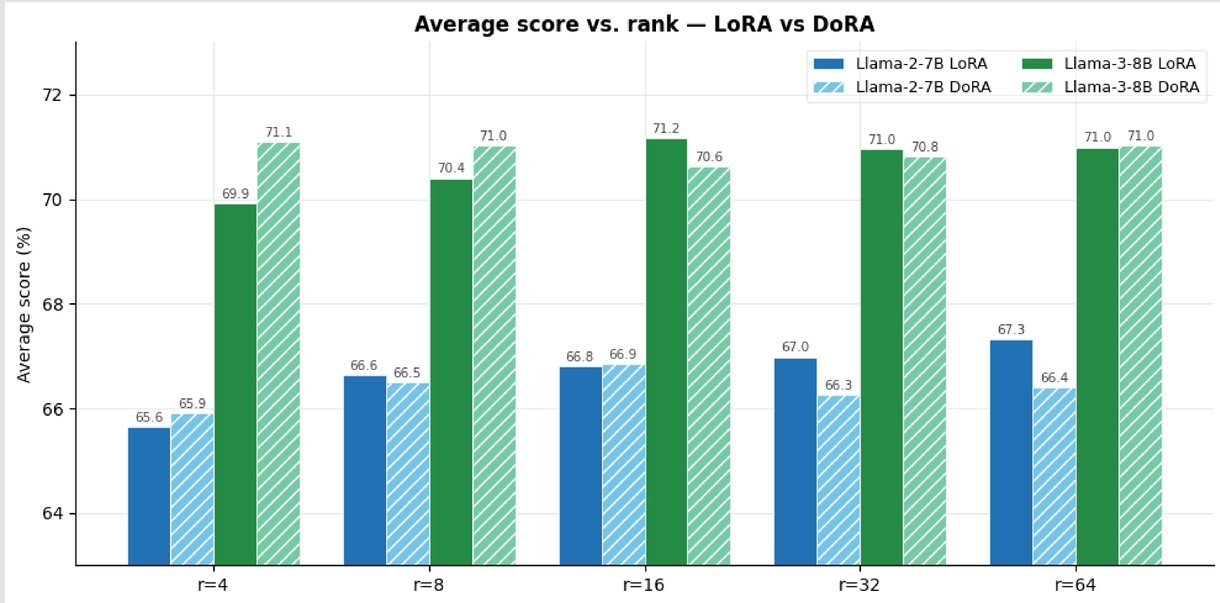
Llama-2-7B and Llama-3-8B

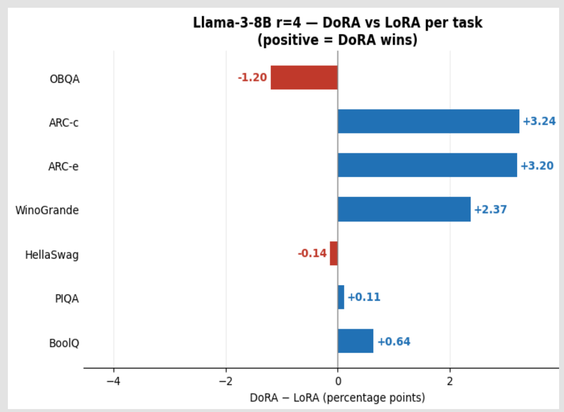
Finetuning

Commonsense Reasoning Benchmark

- BoolQ: yes/no questions from we searches
- PIQA: physical intuition ("what's the best way to...")
- HellaSwag: pick most plausible sentence continuation
- WinoGrande: pronoun resolution requiring common sense
- ARC-e: easy multiple choice science questions
- ARC-c: hard multiple choice science questions
- OBQA: science question requiring reasoning

Results





Outlook

- Train longer: Run at least one full 3-epoch config to confirm the DoRA advantage widens
- Push rank extremes: Add r=1 and r=2

Conclusion

- Llama-3-8B consistently outperforms Llama-2-7B across all ranks and methods scoring ~70–71% average vs. ~65–67%
- DoRA matches LoRA at half the rank on Llama-3-8B: DoRA r=4 (71.09%) beats LoRA r=8 (70.40%), confirming the paper's headline claim
- DoRA is more rank-robust: across r∈{4,...,64}, DoRA's scores span only 0.47 pts on Llama-3 and 0.96 pts on Llama-2, vs. 1.25 and 1.69 pts for LoRA.
- DoRA's advantage is strongest at low rank (r≤8) and shrinks at r≥32, where LoRA catches up — consistent with the original paper.
- Llama-2-7B shows a muted version of the same effect, likely due to undertraining (1,000 steps ≈ 10% of one epoch vs. the paper's 3 full epochs).

References

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